BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 104 · Sound and Compression

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does a bitmap represent pixel colours?

Q2. Why can heavy virtual-memory use slow a program?
A. RAM becomes permanent storage
B. The CPU loses every core
C. All instructions are deleted
D. Secondary storage is slower

Q3. “A parameter is the final printed output of every program.” Correct this. Explain why.

Q4. A microphone records digital sound. “Sampling stores every instant of the sound wave.” Correct this. How does it affect recorded detail?

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I Do · Read new knowledge

Key vocabulary

Sampling — Measuring an analogue signal at regular time intervals.

Sample rate — Number of samples recorded each second.

Bit depth — Number of bits per sample.

Lossless compression — Reduces size without losing original information.

Lossy compression — Removes some information irreversibly.

New learning

Analogue to digital: An analogue sound wave is sampled at regular time intervals, and each sample is quantised to an available binary level.

Sample rate: Sample rate counts samples taken per second. More samples can capture changes in the wave more accurately.

Bit depth and size: Bit depth controls possible amplitude levels. Uncompressed mono size in bits = sample rate × duration × bit depth.

Compression: Lossless methods reconstruct original data exactly; lossy methods discard some information to achieve smaller files.

Worked example

Calculate the uncompressed size of a 10-second mono recording sampled at 8000 Hz with 8-bit depth. [4 marks]

1. Identify three factors

2. 8000 samples/second, 10 seconds, 8 bits/sample.

3. Total samples = 8000 × 10 = 80,000.

4. Size = 80,000 × 8 = 640,000 bits = 80,000 bytes. [4]

We Do · Practise with guidance

Guided 1

Calculate the uncompressed size of 5 seconds of mono audio at 16,000 samples/s and 8 bits/sample. [4]

Worked method: 16,000×5=80,000 samples [1]; ×8=640,000 bits [2]; ÷8=80,000 bytes [1].

Guided 2

Explain the effect of increasing sampling rate. [4]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Each pixel has a colour code stored in binary.

Q2. D — Secondary storage is slower

Q3. A named input received by a subprogram when called with an argument.

Q4. Measure amplitude at regular intervals. Store each rounded (quantised) value in binary. Sample rate and bit depth limit the detail captured.

I Do / We Do pages · Guided practice

Q1. 16,000×5=80,000 samples [1]; ×8=640,000 bits [2]; ÷8=80,000 bytes [1].

Q2. More samples each second [1] can capture the original sound wave more accurately [1], but increases uncompressed data stored per second [1] and therefore file size [1].

Q3. Lossless preserves all original information [1] and can restore identical sound data [1]; lossy removes some information [1] for potentially greater size reduction but cannot restore exact original [1].

You Do pages · Code First

```
rate = 800
seconds = 2
bitDepth = 8
bits = rate * seconds * bitDepth
print(bits)
```

Expected result: 12800

You Do pages · Independent questions

Q1. 4000×4=16,000 samples [1]; ×8=128,000 bits [2]; ÷8=16,000 bytes [1].

Q2. More amplitude levels [1] can improve quantisation precision [1], while increasing uncompressed file size [1].

Q3. Lossless preserves all original data [1], so the original can be reconstructed exactly [1], avoiding irreversible quality loss [1].

Worked method: More samples each second [1] can capture the original sound wave more accurately [1], but increases uncompressed data stored per second [1] and therefore file size [1].

Guided 3

Compare lossy and lossless compression for an audio archive. [4]

Worked method: Lossless preserves all original information [1] and can restore identical sound data [1]; lossy removes some information [1] for potentially greater size reduction but cannot restore exact original [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the number of bits in a mono uncompressed recording.

```
rate = 800
seconds = 2
bitDepth = 8
bits = ...
print(bits)
```

write

Question 1

A mono recording lasts 4 seconds, with 4000 samples/second and 8 bits/sample. Calculate the uncompressed size in bytes. [4] Source: Original OCR J277-style practice question.

Question 2

Explain one effect of increasing bit depth while sample rate and duration stay the same. [3] Source: Original OCR J277-style practice question.

Question 3

Explain why lossless compression is more suitable than lossy compression for an original master recording. [3] Source: Original OCR J277-style practice question.

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Exit Ticket · Answer before checking

A trace table can help check how values change during a program.